

TriWorldBench: A Tri-View Consistency Perspective on Embodied World Models

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Abstract

Embodied world models predict the outcomes of robot actions to support learning and planning. For robots equipped with head and wrist cameras, this requires complementary views: the head view captures the overall task, while wrist views reveal local gripper-object interactions. However, evaluating these views independently cannot determine whether they describe the same action and object state. We introduce TRIWORLDBENCH, a benchmark for evaluating embodied world models through synchronized head, left-wrist, and right-wrist videos. It contains 500 episodes across 50 bimanual manipulation tasks and uses 19 metrics to assess tri-view consistency, task alignment, physical and 3D coherence, motion quality, temporal consistency, and visual quality. By combining cross-view checks with measurements tailored to each camera, the benchmark evaluates whether plausible individual videos also form a consistent prediction of the intended task. We summarize overall performance with TWB-Score and retain per-view results to identify where predictions fail. This extends world-model evaluation beyond single-view visual quality. Code, data, and metric definitions are available at <https://github.com/TriWorldBench/TriWorldBench>.

1. Introduction

A world model predicts how an environment evolves and can support robot learning and planning [1, 8, 26]. Action-conditioned video models make these predictions available as synthetic training trajectories, policy rollouts, or visual subgoals [7]. These uses require more than convincing images: a predicted grasp that looks plausible but violates object contact can provide misleading supervision or guide a planner toward an infeasible action. Evaluation must therefore connect visual quality to the manipulation event a video claims to predict.

The camera setup changes what this evaluation must establish. In the head-and-wrist configuration of Figure 1, the head camera provides workspace context and the wrist cameras reveal local geometry and gripper-object contact [9, 14].

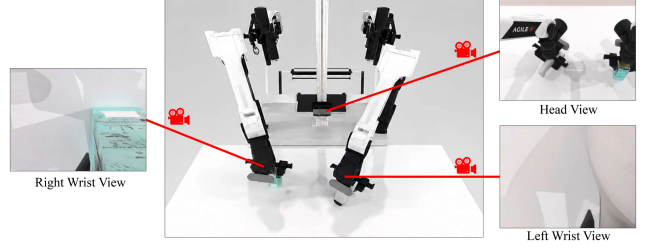


Figure 1. Complementary views of one manipulation event. The head camera provides workspace context; the wrist cameras reveal local geometry and gripper-object contact. TRIWORLDBENCH evaluates both the quality of each predicted view and agreement across the three views.

Their evidence is complementary, not interchangeable: a slipped grasp may be visible only at a wrist, while a close-up may hide which object or arm the task requires. Recent world models predict multiple camera streams jointly [7, 19]. Yet even three individually plausible videos may disagree about whether an object is held, released, or transferred. Single-view quality cannot establish this shared state [18, 23]. Conversely, different appearances do not necessarily imply inconsistency: wide camera baselines, wrist motion, and occlusion naturally change what is visible. The relevant question is whether all views are compatible with the same event, not whether they look alike.

TRIWORLDBENCH makes this distinction explicit by treating a synchronized triplet as one prediction. Building on embodied world-model evaluation [15, 21, 22, 27], it asks three complementary questions: is each view plausible, do the views agree, and does the prediction match the task? Cross-view consistency is a dedicated dimension, not a substitute for task or visual quality. Metrics use the cameras best suited to their evidence, and reference action phases distinguish expected stillness from missing motion. The contribution is thus to assess the relationship between views while preserving the different information each camera provides. This helps distinguish predictions that merely look coherent from those whose views jointly support the intended robot behavior. Table 1 summarizes the 19 metrics and the camera views used to compute each score.

Table 1. The 19 metrics reported by TRIWORLDBENCH, grouped into six dimensions. H, L, and R denote the head, left-wrist, and right-wrist cameras; “H + wrist” denotes a comparison between the head view and the wrist view of the acting arm; “joint” denotes a measurement over the triplet as a whole. H/L/R metrics are computed separately in each view before aggregation. Pixel and structural comparisons use the corresponding ground-truth camera, not a different viewpoint.

Dimension	Metric	What it measures	Cameras
Tri-view consistency	Normalized PSNR	Pixel-level agreement with the corresponding ground-truth camera	H/L/R
	SSIM	Structural similarity to the corresponding ground-truth camera	H/L/R
	VLM Consistency I	Compatibility-first head-to-wrist check of robot state, phase, and object state; occluded evidence tolerated	H + wrist
	VLM Consistency II	Verification-first check: criteria not verifiable from both views receive a deduction	H + wrist
	VLM Consistency III	Exemplar-calibrated against labeled cases of blur, wrong held object, and wrong acting arm	H + wrist
Task alignment	VQA Consistency	Accuracy on a question bank frozen in advance from the reference triplet and phases	joint
	Instruction Following	Whether the rollout uses the required arm and object and reaches the goal state	H
	Semantic Alignment	Similarity between generated and reference head-video captions in text embedding space	H
Physical and 3D coherence	JEPA Similarity	Feature similarity to the reference triplet under a frozen video encoder, camera order preserved	joint
	Interaction Quality	Plausibility of contact, grasp stability, object response, and freedom from penetration	H/L/R
Motion quality	Perspective	Depth-consistent scale, occlusion ordering, shape stability, and camera geometry	H/L/R
	State Alignment	Agreement with the moving or static pattern expected for each view in the reference phase	H/L/R
	Flow Score	Optical-flow magnitude [24], indicating visible motion in each view	H/L/R
Temporal consistency	Trajectory Accuracy	Agreement of predicted and reference end-effector paths under time warping	H
	Subject Consistency	Stability of robot and object identity and appearance across frames	H/L/R
	Background Consistency	Stability of background features across frames	H/L/R
Visual quality	Photometric Smoothness	Motion-aligned appearance stability; penalizes flicker and texture drift	H/L/R
	Image Quality	Frame clarity: blur, noise, exposure, compression [13]	H/L/R
	Aesthetic Quality	Frame-level visual appeal, reported separately from task correctness	H/L/R
			H/L/R

2. Related Work

Embodied world models. World models support robot learning and planning [1, 26]. EnerVerse-AC, Genie Envioner, DreamDojo, and Motus predict action-conditioned robot video [2, 6, 12, 16]. PAVXploreRL further uses physical, action, and visual criteria as post-training rewards [25]. As predictions become training data and decision inputs, evaluation must assess task behavior and agreement across cameras.

Multi-view prediction and consistency. Multi-view models explicitly represent camera relationships. Ctrl-World predicts external and wrist views jointly [7], geometry-aware video generation aligns views through point maps [19], and ReViWo learns representations robust to camera disturbance [20]. In image and video generation, shared priors, 3D-aware attention, and geometric distillation promote geometric consistency [17, 18, 23]. TRIWORLDBENCH tests whether such predictions describe compatible robot and object states despite motion and occlusion.

Video and world-model benchmarks. Benchmarks distinguish visual fidelity from useful prediction. VBench evaluates perceptual and temporal dimensions [10]; WorldScore evaluates controllability, quality, and dynamics [5]. WorldModelBench tests instruction following and physical laws [15], while EWMBench separates scene consistency, motion correctness, and semantic alignment [27]. WorldSimBench combines human-aligned perceptual judgments with action-level evaluation [21]. WorldArena further connects

perceptual assessment to functional utility, evaluating world models as data engines, policy evaluators, and action planners [22]. This motivates separating visual appeal from task alignment. RBench evaluates robot-oriented task correctness and physical plausibility [4], while RoboWM-Bench tests whether generated behaviors translate into executable manipulation [11]. TRIWORLDBENCH adapts these dimensions to complementary camera roles and explicitly evaluates agreement between synchronized views.

3. Benchmark Overview

Task and data. Each episode provides three synchronized initial frames, a natural-language instruction, and an action sequence. A model predicts a head-view video and two wrist-view videos covering the same time span; corresponding timestamps denote the same instant of the task. The test set contains 500 episodes spanning the 50 bimanual manipulation tasks of RoboTwin 2.0 [3], including pick-and-place, stacking, pouring, handover, switch actuation, and tool use. Tasks are recorded under clean and domain-randomized background conditions, introducing appearance variation alongside task diversity. A separate 100-episode validation split with reference videos and state annotations supports local development.¹

Reference-derived action states. Reference states provide a shared temporal context for evaluation. Fixed rules over gripper opening and end-effector displacement segment

¹<https://huggingface.co/datasets/TriWorldBench/Dataset>

each reference trajectory into coarse phases, including idle, approach, grasp, hold, manipulation, release, and completion. The annotations identify the acting arm and support the expected moving or static pattern for each view. They also select phase-level keyframes and the active head-to-wrist pair (Figure 2). These are heuristic action labels, not verified task-success labels. Derived only from the reference trajectory, the boundaries remain fixed across candidate models, so a model’s own failure cannot redefine the phase against which it is judged.

4. Evaluation Protocol

The 19 metrics in Table 1 separate within-view quality from cross-view agreement and task alignment. The protocol combines reference-based comparisons with judgments of the generated videos themselves. Four design choices adapt these measurements to synchronized robot cameras.

Tri-view world consistency. Consistency is assessed both indirectly through the ground truth and directly across views. Normalized PSNR and SSIM compare each generated stream with its corresponding ground-truth stream. Because the reference triplet is synchronized and internally consistent, proximity to it provides an indirect consistency signal. It is not proof of agreement, however, and a valid alternative execution may differ from the reference. Complementary vision-language model (VLM) judgments compare the head and active wrist views. Three rubrics distinguish compatibility despite occlusion, explicit verification in both views, and calibration against labeled examples. Reporting them separately makes their different treatment of uncertain evidence explicit. Visual question answering (VQA) further tests the triplet using questions fixed from the reference and its phases. Together, these checks ask whether the views describe compatible robot and object states without requiring identical appearance across cameras.

Camera-specific evidence. Camera selection follows what a metric needs to observe. Instruction Following uses the head view for task context; Semantic Alignment compares VLM-generated descriptions of the generated and reference head videos in text embedding space. Trajectory Accuracy also uses the head view, where arm paths are more reliably visible than in wrist close-ups. By contrast, JEPA Similarity compares the generated and reference triplets through a frozen video encoder with camera order preserved. Interaction Quality and Perspective are evaluated separately in all three views, capturing local contact and geometry as well as the wider scene. Per-view scores retain these differences instead of treating every camera as an equally reliable observer of every property. For example, a wrist view can reveal a slipping grasp despite plausible head-view object motion.

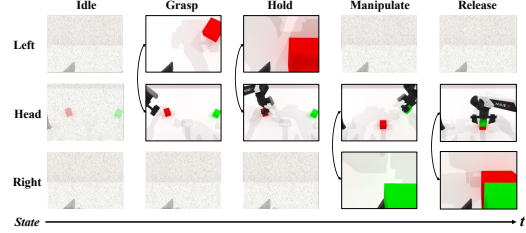


Figure 2. Reference-derived action phases select synchronized keyframes and head-to-wrist pairs (arcs). They also provide context for evaluating the expected motion in each view.

State-conditioned motion. Temporal stability is meaningful only relative to expected motion. State Alignment checks each view’s moving or static pattern; Flow Score measures optical-motion magnitude; Trajectory Accuracy compares head-view end-effector paths with the reference under time warping. More motion is not necessarily more correct motion. Phase-conditioned penalties also adjust subject, background, and photometric consistency when expected motion is missing or unexpected motion appears. This discounts frozen predictions during active phases while preserving legitimate stillness.

Task-grounded visual quality. Visual appeal and task correctness are complementary, not interchangeable. Image and aesthetic scores are therefore accompanied by penalty-adjusted versions that discount quality when head-view trajectory accuracy or cross-view consistency is low. This reduces credit for attractive but unreliable predictions while keeping raw visual quality separate from the task and consistency evidence used to qualify it.

Aggregation. Metrics are averaged within each dimension, and the six dimension scores are combined into TWB-Score. Dimension-level, per-view, head-to-wrist, and joint scores remain available: the aggregate summarizes performance, while the detailed scores help distinguish poor appearance, task mismatch, and cross-view conflict.

5. Conclusion

TRIWORLDBENCH evaluates triple-view world models through the event their videos jointly describe. Its central distinction is between a view that looks plausible, views that agree, and a prediction that matches the task. Camera-aware measurement and reference-derived phases make these questions explicit across 19 metrics, with TWB-Score providing a compact summary. The resulting framework makes cross-view consistency measurable without reducing world-model quality to consistency alone. It supports targeted diagnosis of failures in triple-view predictions for robot learning and planning.

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